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Software Requirements Specification

on

SMART TOURISM MANAGEMENT SYSTEM

JATRAPATH

SEE THE WORLD, WE'LL HANDLE THE REST.

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1.INTRODUCTION

The Smart Tourism Management System is a web-based application designed to provide travelers with an integrated platform for planning and booking trips. The system allows users to explore destinations, calculate travel budgets, book hotels and transportation, redeem gift cards, convert currencies, and access safety information. This system also provides an administrative panel for managing destinations, hotels, transportation services, bookings, users, and contact messages. The goal of this project is to create a centralized tourism solution that simplifies trip planning while ensuring safety, transparency, and convenience.

1.1 PURPOSE

The purpose of this project is to develop a web-based Tourism Management System that:

- Provides detailed information about travel destinations.
- Allows users to book hotels and transportation.
- Calculates estimated travel costs using a budget calculator.
- Displays travel safety scores and emergency contact information.
- Offers gift card purchasing and redemption.
- Includes a currency converter for international travelers.
- Enables administrators to manage users, bookings, destinations, hotels, and transport.
- This document describes the system requirements, functionalities, and overall design of the proposed system.

1.2 SCOPE

The Smart Tourism Management System will:

- Allow users to register and log in securely.
- Enable browsing of destinations and hidden gems.
- Provide booking functionality for hotels and transportation.
- Integrate transportation selection within the booking process.
- Include a budget calculator to estimate total trip cost.
- Display safety ratings and emergency contact information.
- Provide gift card generation and redemption features.

- Offer currency conversion functionality.
- Allow administrators to manage all system data through an admin panel.

The system will operate as a web-based application and store data in a centralized database.

It will not include:

- Real-time online payment gateway (demo payment only)
- AI-based recommendation engine (future enhancement)

1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Here are key terms used in this document:

- **User** – A registered traveler who uses the system.
- **Admin** – System administrator who manages system data.
- **Booking** – Reservation of destination, hotel, and transportation.
- **Gift Card** – Prepaid voucher used during booking.
- **Safety Score** – Rating that indicates destination safety level.
- **FK (Foreign Key)** – Database key that links two entities.
- **PK (Primary Key)** – Unique identifier for an entity.
- **SRS** – Software Requirements Specification.
- **UI** – User Interface.
- **DBMS** – Database Management System.

1.4 REFERENCES

- Software Engineering concepts and SRS documentation standards.
- Database design principles for ER modeling.
- Web development technologies (HTML, CSS, JavaScript).
- Backend development technologies (PHP / Python).
- MySQL Database documentation.

1.5 OVERVIEW

This document provides a detailed description of the Smart Tourism Management System, including:

- Overall system architecture
- User and Admin modules
- Functional requirements
- Non-functional requirements
- ER diagram and database structure
- System constraints and future enhancements

The following sections will describe the system design, database relationships, and implementation approach in detail.

2. OVERALL DESCRIPTION

The system consists of two main actors:

- User (Traveler)
- Admin (System Administrator)

2.1. PRODUCT PERSPECTIVE

The Smart Tourism Management System is a standalone web-based system developed using a three-tier architecture:

- Presentation Layer (Frontend)
- Application Layer (Backend)
- Database Layer (MySQL)

The system operates using client-server architecture:

User Browser → Web Server → Application Logic → Database

The Admin panel is integrated with role-based access control.

2.2. PRODUCT FEATURES

The Smart Tourism Management System shall provide the following major functions:

- User Authentication
- Destination Management
- Booking Management
- Budget Calculation
- Gift Card Management
- Currency Conversion
- Contact Management
- Administrative Management

2.3. USER CHARACTERISTICS

General Users (Travelers)

- Basic internet knowledge
- Able to fill online forms
- No technical expertise required

Admin

- Moderate technical knowledge
- Responsible for system data management
- Authorized to manage bookings, destinations, hotels, and users.

2.4. GENERAL CONSTRAINTS

- The system must run on a web server (XAMPP or hosting server).
- Requires MySQL database.
- Requires internet connection.
- Demo payment system only.
- Must support major web browsers

2.5. ASSUMPTIONS AND DEPENDENCIES

Assumptions:

- Users have internet access.
- Web browser is available.

- Database server is operational.

Dependencies:

- Web hosting environment
- Backend framework (PHP / Flask)
- MySQL DBMS

3. SUPPORTING INFORMATION

3.1. TECHNOLOGIES USED

The Smart Tourism Management System will be developed using modern web technologies to ensure reliability, scalability, and performance.

The following technologies will be used:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP / Node.js
- Database: MySQL
- Web Server: Apache / XAMPP / Node Server
- Development Tools: Visual Studio Code
- Design Tool: draw.io (for ER)

These technologies are selected to provide a responsive user interface and efficient data management.

3.2. EXTERNAL INTERFACES

This section describes how the system interacts with external components.

- **USER INTERFACE**

- The system will provide a web-based interface accessible through standard web browsers.
- The interface will include pages such as Home, Destinations, Booking, Budget Calculator, Login/Register, and Admin Panel.
- The system will support responsive design for desktop and mobile devices.

- **HARDWARE INTERFACE**

- The system will operate on standard computers and mobile devices.
- No special hardware is required.

- **SOFTWARE INTERFACE**

- The system will interact with a relational database (MySQL).
- It may integrate with payment gateways for gift card purchases (if applicable).

- **COMMUNICATION INTERFACE**

- The system will use HTTP/HTTPS protocol for communication between client and server.
- Internet connectivity is required for accessing the system.

3.3. NON-FUNCTIONAL REQUIREMENTS

This section describes system quality attributes.

- **PERFORMANCE REQUIREMENTS**

- The system shall respond to user requests within 3 seconds under normal load.
- The system shall support multiple concurrent users.

- **SECURITY REQUIREMENTS**

- User authentication shall be required for booking and administrative access.
- Passwords shall be stored securely using encryption.
- Only authorized administrators shall access the Admin Panel.

- **USABILITY REQUIREMENTS**

- The system shall provide a user-friendly interface.
- Navigation shall be simple and intuitive. The system shall provide clear error messages.

- **RELIABILITY REQUIREMENTS**
 - The system shall ensure accurate booking and transaction processing.
 - Data shall be stored securely without loss.
- **AVAILABILITY REQUIREMENTS**
 - The system shall be available 24/7 except during maintenance.
- **MAINTAINABILITY REQUIREMENTS**
 - The system shall allow easy updates to destinations, hotels, and transportation data.
 - The code shall be modular and structured.

3.4. SYSTEM ARCHITECTURE OVERVIEW

The Smart Tourism Management System will follow a three-tier architecture:

1. Presentation Layer

The Presentation Layer is responsible for handling all interactions between the user and the system. It provides the graphical user interface (GUI) that allows users and administrators to access system functionalities through a web browser.

For the Smart Tourism Management System, this layer includes:

- Home Page displaying featured destinations
- Destination browsing page
- Destination details page showing safety score and emergency contact information
- Booking page for selecting hotel and transportation
- Budget calculator interface
- Gift card purchase and redemption page
- Currency converter interface
- Login and registration pages
- User dashboard
- Admin panel

This layer collects user inputs such as:

- Registration details
- Login credentials
- Booking information (travel date, number of persons)
- Gift card codes
- Contact messages

It also displays system responses such as:

- Destination information
- Booking confirmations
- Budget estimates
- Error messages
- Administrative updates

The Presentation Layer does not contain business logic and does not directly interact with the database. All user requests are forwarded to the Application Layer for processing.

2. Application Layer

The Application Layer contains the core business logic of the system. It processes the requests received from the Presentation Layer and performs necessary validations and calculations before interacting with the database.

In the Smart Tourism Management System, this layer performs the following operations:

- Verifies user authentication and authorization
- Determines whether a user is a traveler or an administrator based on role Processes booking requests and validates selected destination, hotel, and transportation.
- Calculates total travel cost using budget calculation logic
- Validates gift card codes and updates gift card status
- Performs currency conversion calculations
- Stores booking records
- Handles contact message submission
- Allows administrative actions such as managing destinations and users

This layer ensures:

- Users cannot access admin functions without proper authorization
- Gift cards cannot be reused once redeemed
- Travel cost calculations are accurate
- Booking data is validated before storage

The Application Layer acts as an intermediary between the user interface and the database.

3. Data Layer

The Data Layer is responsible for managing and storing all persistent data in the system. It uses a relational database management system to maintain structured and secure data storage.

The database stores information related to:

- Users (including role information)
- Destinations and their safety scores
- Hotels associated with destinations
- Transportation options
- Booking records
- Gift card details and status
- Contact messages

This layer ensures:

- Data integrity through primary and foreign key constraints
- Secure storage of user credentials
- Accurate maintenance of booking history
- Reliable management of transactional data

The Data Layer does not interact directly with users. All data operations are performed through the Application Layer.

4. Client-Server Interaction

The system follows a client-server model.

- The client refers to the user's web browser.
- The server hosts the application logic and database.
- Communication between client and server occurs using HTTP/HTTPS protocols.

When a user performs an action (such as booking a destination):

- The browser sends a request to the server.
- The server processes the request in the Application Layer.
- The Application Layer interacts with the database.
- The server sends the processed result back to the browser.

This architecture ensures:

- Efficient request handling
- Secure data processing
- Clear separation of system components
- Improved scalability and maintainability

4. ANALYSIS MODEL

This section describes the graphical models used to represent the structure and behavior of the Smart Tourism Management System.

4.1. USE CASE DIAGRAMS

The Use Case Diagram illustrates the interaction between system actors and system functionalities.

Actors:

1. User (Traveler)
2. Admin

Use Cases for User:

- Register
- Login
- Browse Destinations
- View Destination Details
- View Hidden Gems
- View Safety Score
- Book Travel Package
- Select Hotel
- Select Transportation
- Use Budget Calculator
- Purchase Gift Card
- Redeem Gift Card
- View Booking History
- Send Contact Message

Use Cases for Admin:

- Login
- Manage Users
- Manage Destinations
- Manage Hotels
- Manage Transportation
- Manage Bookings
- Manage Gift Cards
- View and Respond to Contact Messages

The Use Case Diagram visually represents these actors and their interactions with the system.

4.2. ENTITY-RELATIONSHIP DIAGRAMS

The Entity-Relationship Diagram represents the database structure of the system.

Main Entities:

- User
- Admin
- Destination
- Hotel
- Transportation
- Booking
- GiftCard
- Contact_Message

Key Relationships:

- User makes Booking
- Booking belongs to Destination
- Destination has Hotel
- Destination offers Transportation
- User purchases GiftCard
- User sends Contact_Message
- Admin manages system entities

The ERD defines primary keys, foreign keys, and cardinality constraints.

4.3. DATA FLOW DIAGRAMS

Data Flow Diagrams describe how data moves through the system.

Level 0 (Context Diagram)

The system interacts with:

- User
- Admin

Inputs:

- Registration information
- Login credentials
- Booking details
- Gift card information
- Contact messages

Outputs:

- Booking confirmation
- Budget estimation
- Destination information
- Administrative reports

Level 1 DFD Major system processes include:

- User Authentication Process
- Destination Management Process
- Booking Management Process
- Gift Card Processing
- Contact Handling
- Administrative Management

Data Stores include:

- User Database
- Destination Database
- Booking Database
- Gift Card Database
- Contact Message Database

4.4. STATE DIAGRAMS

State Diagrams describe the different states of important system entities.

Booking State Diagram States include:

- Created
- Pending
- Confirmed
- Cancelled
- Completed

Transitions occur based on user or admin actions.

Gift Card State Diagram States include:

- Generated
- Active
- Redeemed
- Expired

Transitions occur when the gift card is used or reaches expiry.

5. APPENDICES

This section includes additional supporting information relevant to the system.

It may contain:

- Glossary of Terms
- Acronyms and Abbreviations
- References
- Future Enhancements
- Sample Screenshots
- Test Cases